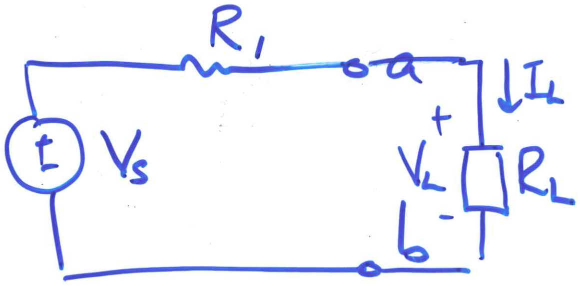
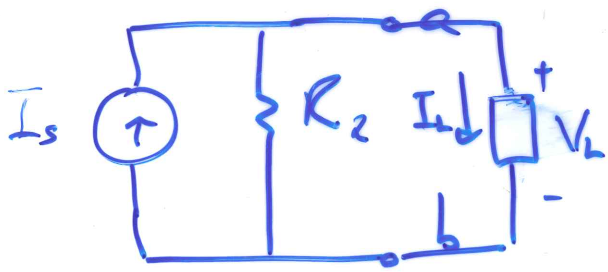
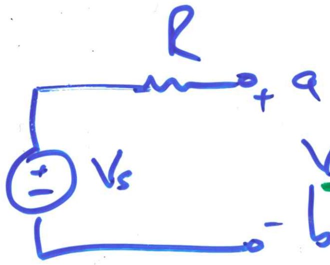
Source Transformation

1. V_{ab} for a-b Open Circuit? ($I_{ab} = 0$)
2. I_{ab} for a-b Short Circuit? ($V_{ab} = 0$)
3. R_{ab} for all sources off?

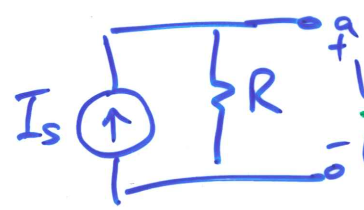




1

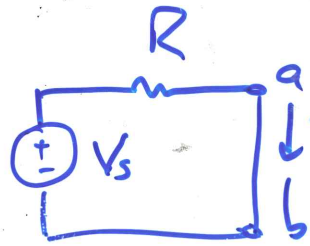


$V_{ab} = V_s$

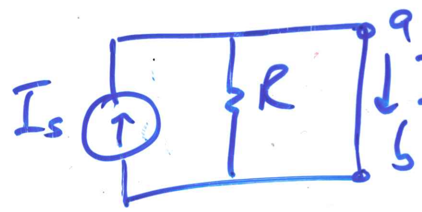


$V_{ab} = I_s R_2$

2

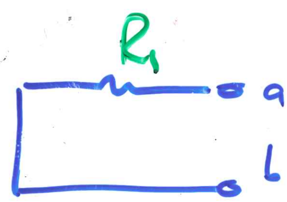


$I_{ab} = V_s / R_1$

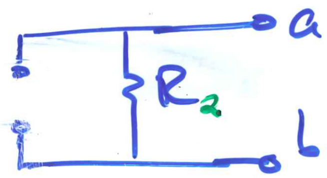


$I_{ab} = I_s$

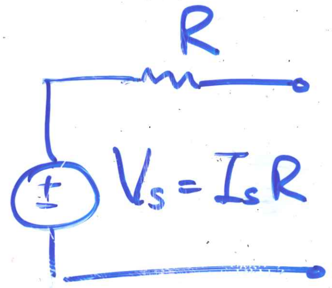
3



$R_{ab} = R_1$

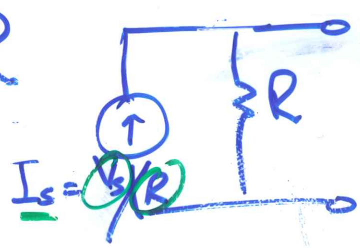


$R_{ab} = R_2$



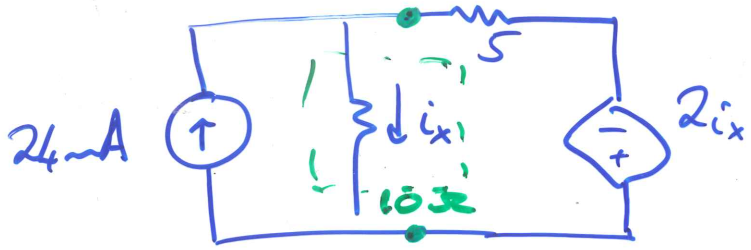
For equivalence:

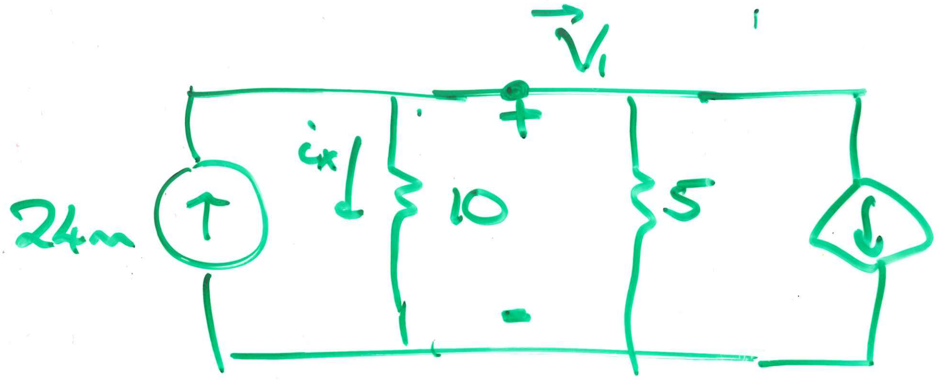
* $R_1 = R_2 = R$
* $V_s = I_s R$



PP4.7

$(i_x = 7.06\,mA)$.





$V_s = 2i_x$

$I_s = 2i_x/5$.

$\rightarrow \quad \sum I_{OUT} = \sum I_{IN}$

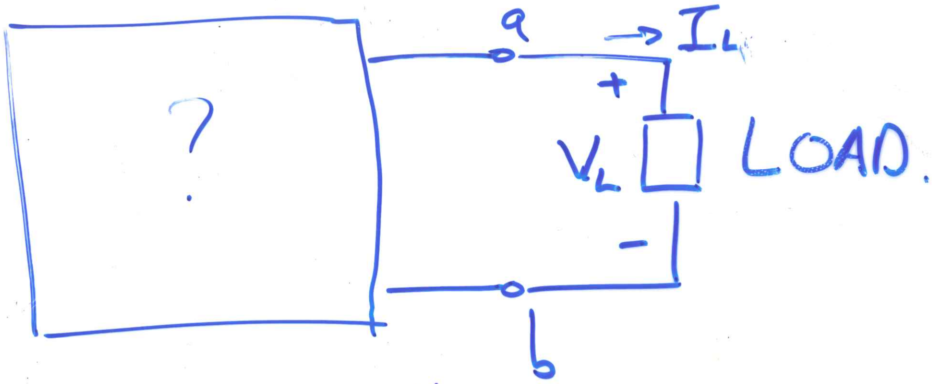
(1) $$\frac{V_1}{10} + \frac{V_1}{5} + \frac{2i_x}{5} = 24\,mA.$$

(2) $$V_1 = 10\,i_x$$

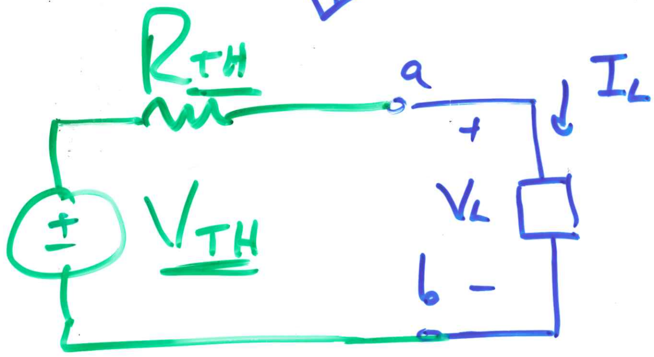
$$\frac{10i_x}{10} + \frac{10i_x}{5} + \frac{2i_x}{5} = 24\,mA.$$

$$i_x = ?$$

Thevenin / Norton

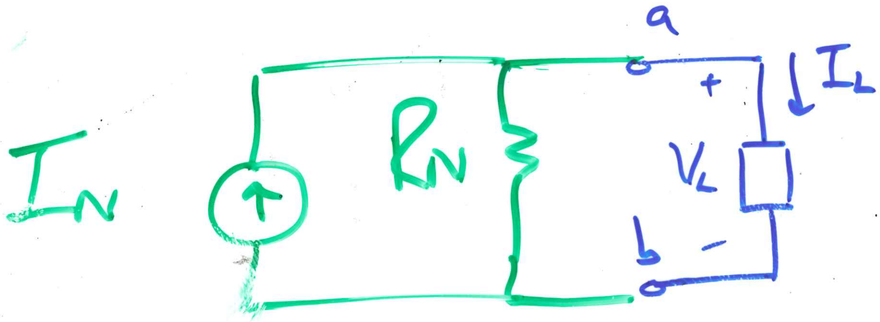


$\Downarrow$

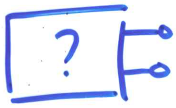


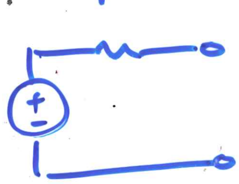
Thevenin equivalent.

$\Downarrow$

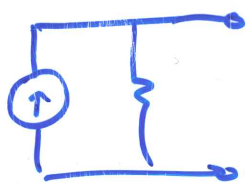


Norton equivalent.

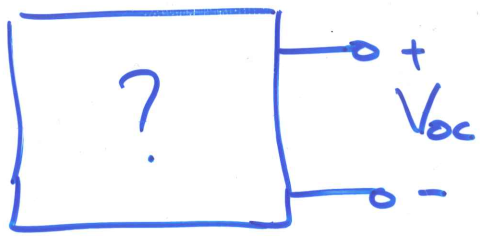
We can replace  with:



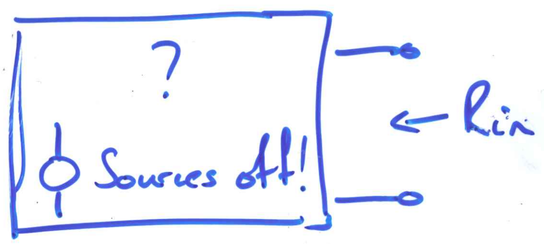
or



Thevenin:

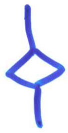


① V_{TH} is open circuit voltage of 

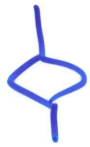


② R_{TH} is resistance of  if sources are off.
(Keep  sources on!)

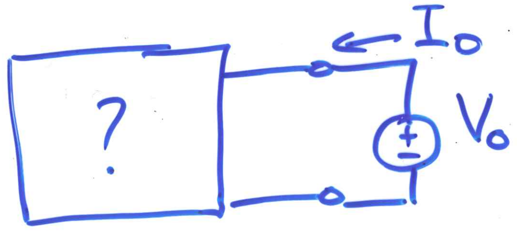
R_{TH}

① No  in circuit

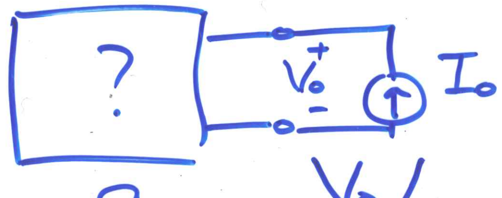
* Use series and $\parallel$ theory to simplify circuit to one resistor.

②  is in circuit

* Simplify series + $\parallel$ if possible.
* Apply test source



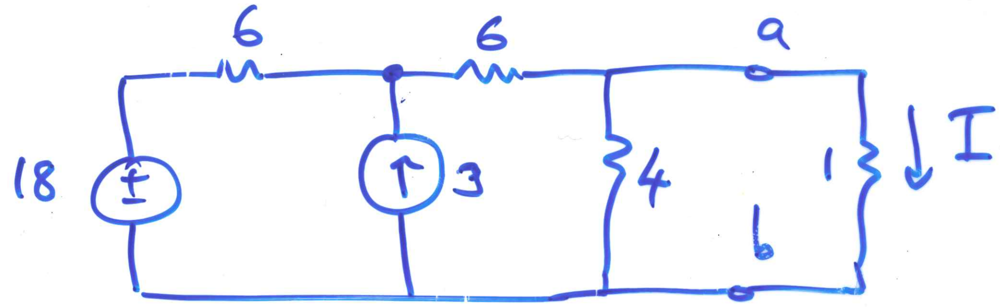
or



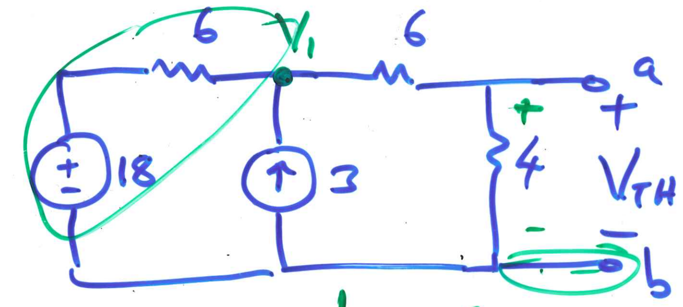
$\therefore R_{TH} = V_o / I_o$.

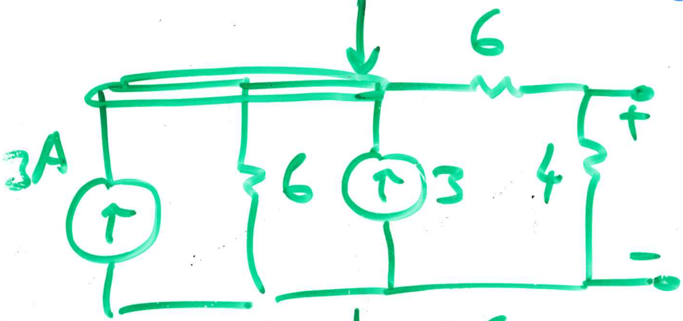
PP 4.8: a) Determine V_{TH} and R_{TH} for the circuit to the left of a-b.

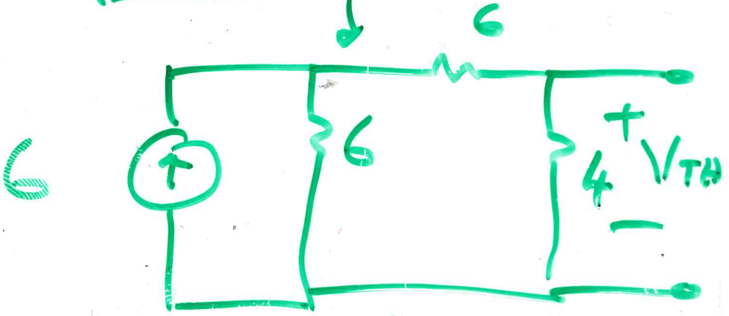
b) Determine I.

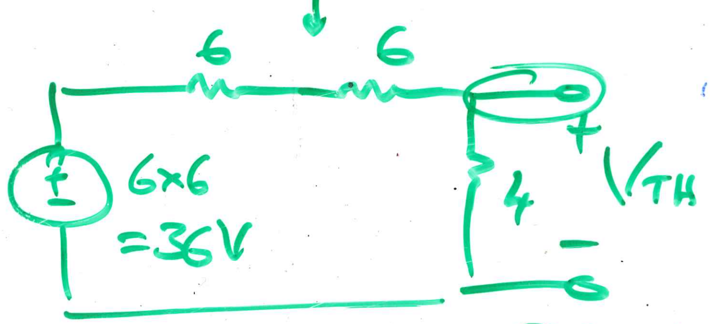


① V_{TH}.



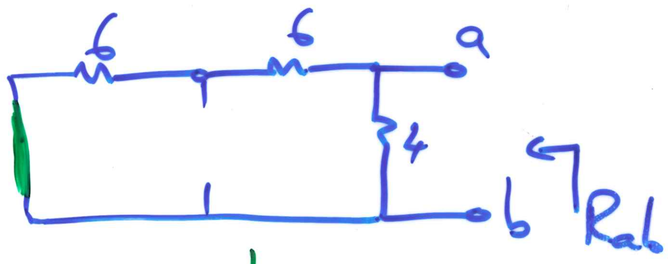


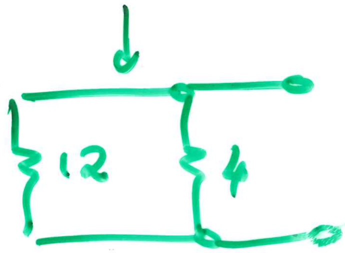




$$V_{TH} = \frac{36 \times 4}{4+6+6} = \cancel{8.3\ V}\ 9V$$

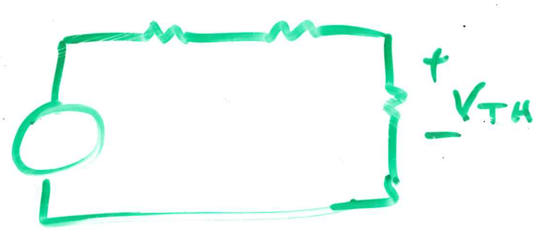
② R_{TH}



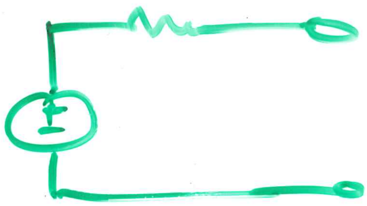


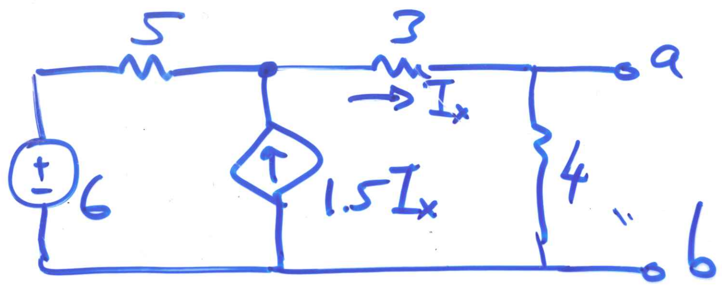
$R_{||} = 12 || 4$

$$R_{TH} = \frac{12 \times 4}{16} = 3\ \Omega$$



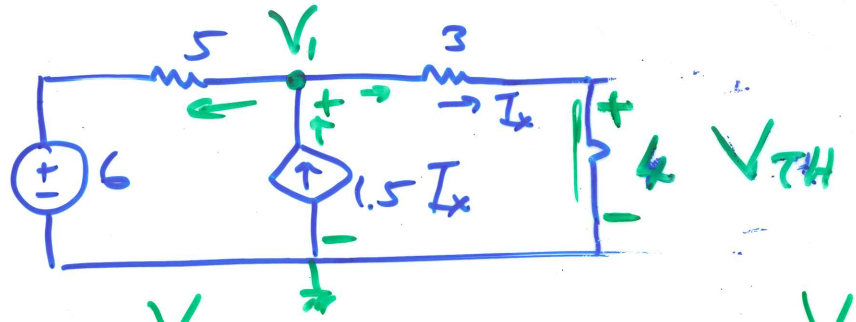
PP 4 - 9





Find Thev equivalent circuit.

① V_{TH}



$*\quad I_x = \frac{V_1}{3+4}$

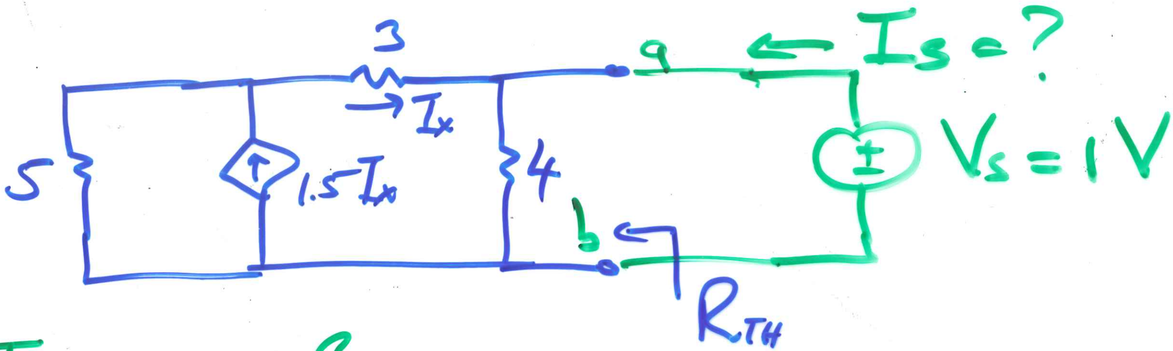
$*\quad V_{TH} = \frac{V_1 \times 4}{3+4} = \underline{5.3V}$

$*\quad \frac{V_1 - 6}{5} + \frac{V_1 - 0}{7} = 1.5 I_x$

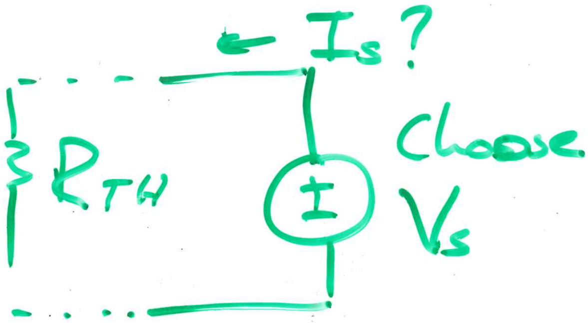
$$\frac{V_1}{5} - \frac{6}{5} + \frac{V_1}{7} = \frac{3}{2}\left(\frac{V_1}{7}\right).$$

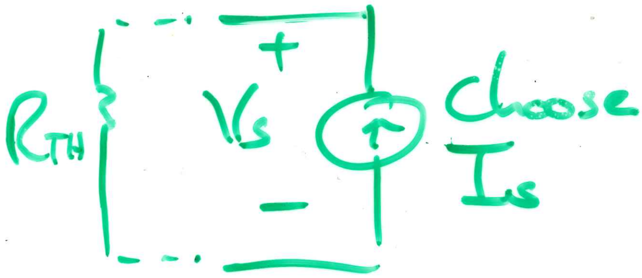
$$V_1 = \underline{9.3V}$$

② R_{TH}



Test Source.





$$R = V_s / I_s$$

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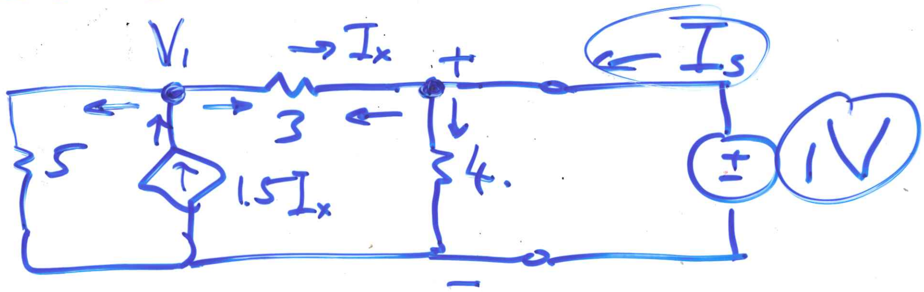
* Test Source: $V_s = 1V$ $I_s = ?$



* $\frac{V_1}{5} + \frac{V_1 - 1}{3} = 1.5 I_x$ ①

* $I_x = \frac{V_1 - 1}{3}$ ②

$V_1 = -5V$ (from ① and ②).

* $I_s = \frac{1}{4} + \frac{1 - V_1}{3}$

$I_s = \underline{2.25A}$

$$R_{TH} = 1/I_s = 1/2.25 = 0.4\ \Omega$$